

Lab 1 — Practical Introduction to Firewall Deployment Using GNS3

➤ Learning Objectives

By the end of this lab, students will be able to:

- Deploy a firewall inside a simple network topology
 - Understand LAN/WAN segmentation
 - Configure basic network interfaces
 - Create and test firewall filtering rules
 - Capture and analyze allowed and blocked traffic
 - Interpret firewall logs
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➤ TECHNICAL PREREQUISITES

Hardware Requirements

- Computer with minimum 8 GB RAM (16 GB recommended)
- At least 25 GB free disk space
- Hardware virtualization enabled in BIOS (Intel VT-x / AMD-V)

Required Software

1. GNS3 (latest stable version)
<https://www.gns3.com/software/download>
2. GNS3 VM (OVA file) – mandatory for performance
(Provided during installation)
3. VirtualBox or VMware Workstation
4. pfSense Community Edition ISO
<https://www.pfsense.org/download/>
5. Wireshark (recommended for packet analysis)

Pre-Lab Verification

Before the session, students must:

- Successfully start GNS3 VM
- Verify VM status shows “Connected”
- Create and open a new empty project

Part 1 — Firewall Deployment & Client Configuration

NETWORK TOPOLOGY

[VPCS-LAN] ---- [pfSense Firewall] ---- [NAT / Internet]

Components in GNS3:

- 1 VPCS (internal client)
- 1 pfSense VM
- 1 NAT node (simulated Internet)

Step 1 – Import pfSense into GNS3

- Create a new VM template
- Select pfSense ISO
- Allocate at least 2 network interfaces

Assign:

Adapter 1 → WAN

Adapter 2 → LAN

Step 2 – Install pfSense

Default configuration:

- ZFS filesystem
- Reboot after installation
- Assign interfaces:

WAN → em0

LAN → em1

Step 3 – Configure LAN IP Address

Set:

LAN IP: 192.168.10.1/24

DHCP: Enabled

Inside VPCS: ip dhcp

Verify: show ip

Test connectivity:

ping 192.168.10.1

ping 8.8.8.8

If properly configured → Internet access should work.

Part 2 — Traffic Observation & Firewall Rule Creation

In GNS3:

- Right-click LAN link
- Start capture
- Open in Wireshark

Observe:

- ICMP echo request/reply
- TCP three-way handshake

This visualizes traffic flow and packet inspection.

➤ [Exercise 1 – Block Outgoing ICMP](#)

1. **Navigate to:** Firewall > Rules > LAN
2. **Add rule:**

```
Action: Block
Protocol: ICMP
Source: LAN net
Destination: any
```

Apply changes.

Test: `ping 8.8.8.8`

Expected result: **Failure.**

Check logs: Status > System Logs > Firewall

➤ [Exercise 2 – Allow HTTP but Block HTTPS](#)

Create two rules:

- Pass TCP port 80
- Block TCP port 443

Test using a browser (if using a real LAN client).

➤ [Exercise 3 – Source-Based Filtering](#)

Add a second VPCS client.

- Allow traffic from PC1
- Block traffic from PC2

Understand rule granularity and filtering by source IP.

TIPS & BEST PRACTICES

- ✓ Firewall rules are processed top-down
 - ✓ Always verify local connectivity before Internet tests
 - ✓ If nothing works → check gateway configuration
 - ✓ If DHCP fails → verify LAN DHCP service is enabled
 - ✓ Always inspect logs before troubleshooting blindly
 - ✓ A firewall enforces explicit policies ; nothing is implicit
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REFLECTION QUESTIONS

- What happens if no LAN rule is defined?
- Why does rule order matter in firewall configuration?
- What is the difference between stateful and stateless firewalls?
- Why might blocking ICMP be considered bad practice in real environments?
- Where would you place a DMZ in this topology?
- What happens if the firewall becomes unavailable?

“A network without filtering isn’t a network ; it’s an open invitation.”